DTW+S: Shape-based Comparison of Time-series with Ordered Local Trends — a complete summary

Ajitesh Srivastava (USC). arXiv:2309.03579v3 (Dec. 2024); published at AAAI 2025 as "Aligning time-series by local trends: Applications in public health". arxiv.org/abs/2309.03579

Problem. Standard distances (Euclidean / MAE, correlation, plain DTW) can rank a flat, information-free line as *closer* to a ground-truth curve than a forecast that reproduces the right shape but is slightly off in height or time. The author wants a distance for which two series are similar *iff* they show a *similar sequence of local trends occurring around similar times*, scale-tolerant, and — crucially — *interpretable* for applied users (epidemiologists), who resist black-box metrics. Target setting: series with *ordered local trends* (e.g. epidemics: surge → increase → peak → decrease).

Core idea. Two stages. (1) Transform each series into an interpretable *Shapelet-Space Representation* (SSR): slide a window of length w over the series and, at each position, map the window to a d-dimensional vector of similarities to pre-chosen *shapelets* (shapes of interest). This yields a matrix $A \in \mathbb{R}^{d\times(T-w+1)}$ whose columns are interpretable local trends over time. (2) Apply *DTW* between the SSR matrices of two series, using squared Euclidean distance between columns $\|A[:,i]-B[:,j]\|^2$, so that similar trends are matched even if shifted in time (warping window τ; e.g. $\tau = 5$ weeks for epidemics, a tuned hyper-parameter for classification). Distance = DTW on shapes = **DTW+S**.

Shapelet space & closeness preservation. Shapelets are fixed domain shapes (used throughout: `increase`=[1,2,3,4], `surge`=[1,2,4,8], `peak`=[1,2,2,1], `flat`=[0,0,0,0]). Each coordinate is a (flatness-gated) correlation of the normalized window to a shapelet; an explicit "flatness" factor stops noisy near-flat windows from looking like real shapes. The design enforces *Closeness Preservation*: two windows get similar representations iff one is a scale/translation of the other, or both are "almost flat".

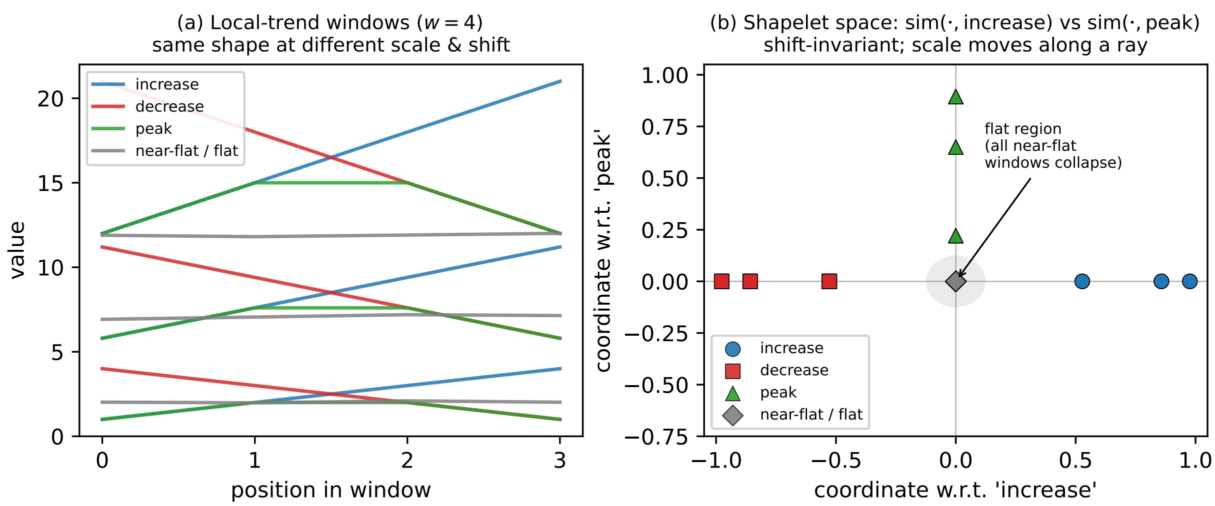

Figure 1: Closeness preservation, computed with the paper's formulas (shapelets `increase`/`peak`; coordinate $\mathrm{sim}(x, s_i) = (1-\gamma)\,\mathrm{corr}(x, s_i)$). **(a)** Length-4 windows of the same shape at different scales and time-shifts. **(b)** Their shapelet-space coordinates: *translation-invariant* (shifted copies overlap), shape sets the *direction* (increases on $+x$, decreases on $-x$, peaks on $+y$), and amplitude moves a window *along a ray*; every "almost-flat" window — whatever its underlying shape — collapses into the shaded flat region at the origin. This is exactly Property 1: similar representation $\iff$ (i) one is a scale/translation of the other, or (ii) both are almost flat.

Theory. (1) *Theorem 1+2:* exactly $w-1$ linearly independent shapelets plus the flat shapelet are *necessary and sufficient* to satisfy closeness preservation. (2) *Theorem 3 (expressive power):* a proper shapelet-space transform can discriminate *any* trend descriptor (including, via the flat dimension, some scale information) — unlike finite-label encoders that collapse distinct shapes (e.g. "peak" vs "increase").

Ensembling (a key contribution). *Setting:* several models each forecast the same epidemic, and we want one "consensus" curve. The usual method (a *mean ensemble*) just averages the curves *vertically* — at each week, take the average of the values. This misleads when the peaks don't line up in time. Example: model A peaks at week 10 and model B at week 14, each reaching 1000 cases. Averaging week-by-week gives no sharp peak — only a lower, wider bump around week 12 that *neither* model predicted, making the wave look far milder than both forecasts say.

DTW+S fixes this by averaging *matching events* instead of fixed clock-times: it first uses the shape alignment to see that A's peak and B's peak are *the same event*, then averages their *timing* (weeks 10 and 14 $\rightarrow$ week 12) *and* their *height* (1000 and 1000 $\rightarrow$ 1000). The consensus curve then keeps a real peak of $\sim$1000 at week 12. *How it is computed* (*DTW+S Barycenter Averaging*, Alg. 2): pick one curve as a starting reference, align every curve to it with DTW+S and average to get an updated reference, and repeat until the reference stops changing.

Clustering & classification. The DTW+S distance plugs into any distance-based method: the paper uses agglomerative clustering (number of clusters via Silhouette) and 1-nearest-neighbor classification (chosen so the distance measure alone decides the label).

Experiments & results. (1) *Ensemble:* on 12 synthetic peak sets, DTW+S BA cuts fractional error in recovered peak *size and timing* to $\approx$0.01–0.03, vs. $\approx$0.1–0.4 for mean / plain-DTW / z-normalized-DTW ensembles. (2) *Clustering:* gives more sensible groupings of epidemic curves than DTW or SSR alone. (3) *Classification (UCR archive, 1-NN):* DTW+S beats plain DTW on a large subset of the 64 datasets where local trends (not scale) are decisive, and beats shapeDTW on nearly half; light pre-smoothing (tuned) further improves many datasets. (4) *Interpretability:* SSR heat-maps directly reveal *why* two series differ (e.g. a bright "peak" dimension around $t = 25$ separating two classes).

Relation to prior work. Closest is shapeDTW (point-wise shape descriptors); DTW+S instead uses *interpretable, domain-chosen* shapelets with provable closeness-preservation guarantees, and is purpose-built for ordered-trend / epidemic applications rather than general use.

Limitations. Not meant for general-purpose tasks where scale dominates shape; best use needs domain knowledge to choose shapelets and set warping/smoothing parameters (Theorems 1–2 only guide the shapelet choice); current DTW step is $O(T^2)$ time and space (speed-ups left to future work). Notably, evaluation is entirely *computational* (ensembling / clustering / classification) — there is *no human-perception study* of whether its alignments match how people judge similarity.